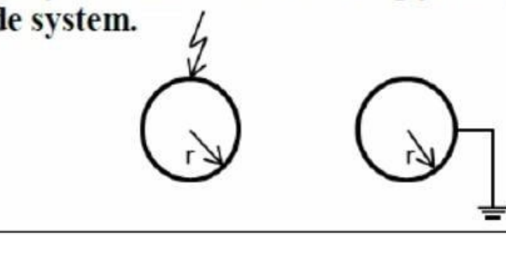
2025 çıkanlar

1)
2) sözel konuşmacı + projeler
3) difference method
4) corona veya refraction gibi sözel soru

Birimleri yazmayı unutma

QUESTION-1 (25 pts): In a **sphere-sphere** electrode system given in the figure below, **60 kV** is applied to one of the electrodes, while the other electrode is grounded. The breakdown strength of air is **30 kV/cm**. If the electrode separation is **0.8 cm**, and **k=0.85** determine the radius of the spheres. Make your calculations **by modifying the concentric spherical electrode system formula accordingly to represent a sphere-sphere electrode system.**



Equation for a concentric spherical electrode system:

$$E_{max} = \frac{U}{\frac{r_1}{r_2}(r_2 - r_1)}$$

$r_2 = r + \frac{d}{2}$ unutma

$E_b = 30 kV/cm$ $d = 0,8 cm$ $k = 0,85$ $V = 60kV$

$$E_b = k\left(\frac{V/2}{\frac{r}{r+\frac{d}{2}}\left(r+\frac{d}{2}-r\right)}\right) = k\left[\frac{(V/2)(2r+d)}{r.d}\right] \quad \frac{kV/cm}{*}$$

$$\cancel{30}\times\cancel{10^3} = 0,85 \times \left[\frac{\left(\frac{\overset{30}{\cancel{60}}}{\cancel{2}}\right)\times\cancel{10^3}\times(2r+(0,8))}{(r)(0,8)}\right] = \frac{V/cm}{*}$$

$$\frac{1}{0,85} = \frac{2r+0,8}{r}$$

$$1,18 = \frac{2r+0,8}{r}$$

$1,18r = 2r + 0,8$

$-0,8 = 0,82r$

~~$\boxed{-1,025 = r}$~~

$\boxed{r = 1,025\ cm}$ *

QUESTION 2:

(25 pts) In a **three phase 380 kV transmission line,** diameter (diameter) of conductors are 70 mm². The breakdown strength of air is **30 kV/cm** and the correction factor is **0.90**. Using **approximated field calculation**, determine **the minimum distance between two phases.**

Equation for a coaxial cylindrical electrode system:

$$E_{max} = \frac{U}{r_1 ln\frac{r_2}{r_1}}$$

$$E_{max} = \frac{k\frac{V}{2}}{r_1 . ln\left(\frac{r+\frac{d}{2}}{r}\right)}$$

$V = 380 \times 10^3 V$
$k = 0,9$
$E_{max} = 30kV/cm$

Diameter $= 70\ mm$
$r = 3,5 cm$

$$E_{max} = \frac{0,9 \times \frac{380}{2}}{3,5 . ln\left(\frac{3,5+d/2}{3,5}\right)} = 30kV/cm$$

$$E_{max} = \frac{48,857}{ln\left(\frac{3,5+d/2}{3,5}\right)} = 30kV/cm$$

$$E_{max} = 1,6286 = ln\left(\frac{3,5+d/2}{3,5}\right)$$

$$\frac{3,5+\frac{d}{2}}{3,5} = e^{1,6286}$$

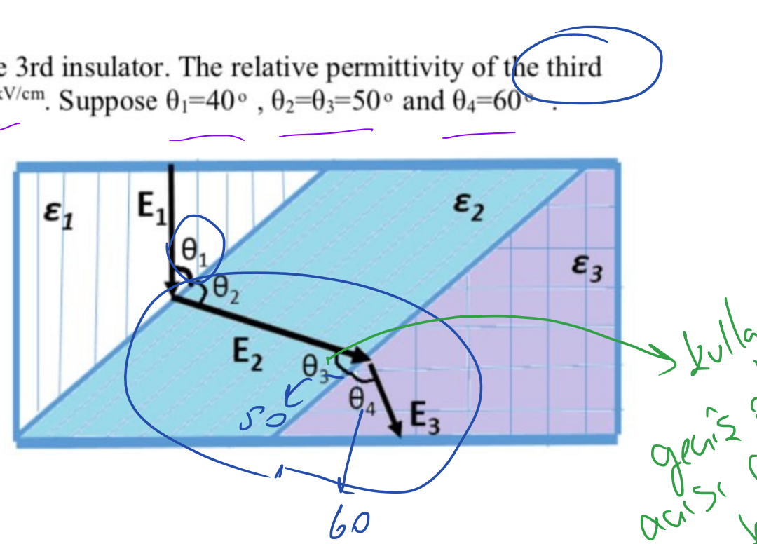
$3,5 + \frac{d}{2} = 5,0967 \times 3,5$

$d = 28,67714\ cm$

Q1) In the following insulation system, there is a discharge in the 3rd insulator. The relative permittivity of the third insulator is $\varepsilon_{r3} = 2.3$ and its electric field strength is $E_{b3} = 90\ ^{kV/cm}$. Suppose $\theta_1 = 40°$, $\theta_2 = \theta_3 = 50°$ and $\theta_4 = 60°$. Determine:

a) The minimum values of E_1 and E_2. **(10 P)**
b) The relative permittivities of ε_{r1} and ε_{r2}. **(10 P)**

($\varepsilon_0 = 8.85 \times 10^{-12}$)



kullanmayız geçiş olmayan açısı çünkü diğer bölgelerin normalle yaptı açı değil

$\varepsilon_{r3} = 2,3$
$E_{b3} = 90 kV/cm$
$\varepsilon_0 =$
$\theta_1 = 40$
$\theta_2 = 50$
$\theta_3 = 50$
$\theta_4 = 60$

tangential

$$E_1 . \sin(\theta_1) = E_2 . \sin(\theta_2) = E_3(\sin\theta_4)$$

$$E_1 . \sin(40) = E_2 . \sin(50) = \underset{90}{E_3} . \sin(60)$$

$E_2 = 101,83\ kV/cm$

$E_1 = 121,27\ kV/cm$

Normal bileşen (D)

$$\varepsilon_1 . E_1 . \cos(\theta_1) = \varepsilon_2 . E_2 . \cos(\theta_2) = \varepsilon_3 . E . \cos(\theta_4)$$

$$\varepsilon_1 = \frac{\varepsilon_3 . E_3 . \cos(\theta_4)}{E_1 . \cos(\theta_1)} = \frac{2,3 . 90 . 0,5}{121,27 . 0,7660} = 1,11$$

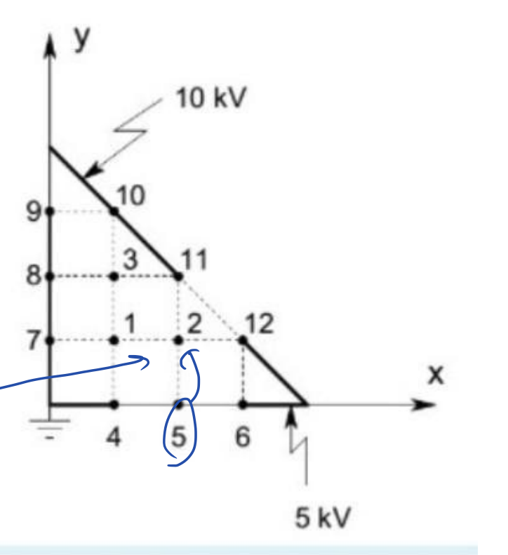
$$\varepsilon_2 = \frac{\varepsilon_3 . E_3 . \cos(\theta_4)}{E_2 . \cos(\theta_2)} = \frac{2,3 . 90 . 0,5}{101,83 . 0,6428}$$

$$\varepsilon_2 = \frac{103,5}{65,49} \approx 1,58$$

Q. Use Finite Difference Method to determine the potentials of free nodes 1, 2 and 3. Where h=4 cm, $V_4 = V_7 = V_8 = V_9 = 0$ Volt, $V_{10} = V_{11} = 10$ kV, and $V_6 = V_{12} = 5$ kV. For node 5, Neumann boundary condition

$$\left(\frac{\partial V}{\partial y}\right)_5 = 0$$

y düzleminde



$$\frac{V_{komşu} - \overset{V_5}{V_{sınır}}}{h} = \frac{V_2 - V_5}{4} = 0$$

$\boxed{V_2 = V_5}$

FDM 'de Laplace Denklemi

$$V_{i,j}^0 = \frac{1}{4}\left(V_{üst} + V_{alt} + V_{sol} + V_{sağ}\right)$$

Node 1

$$V_1 = \frac{1}{4}\left(V_3 + V_4 + V_7 + V_2\right)$$

$$= \frac{1}{4}\left(V_3 + 0 + 0 + V_2\right) = \frac{1}{4}\left(V_3 + V_2\right) \quad ①$$

Node 2

$$V_2 = \frac{1}{4}\left(10 + V_2 + V_1 + 5\right) \Rightarrow 3V_2 = 15 + V_1 \quad ②$$

Node 3

$$V_3 = \frac{1}{4}\left(10 + 0 + 0 + V_1\right) \Rightarrow \frac{1}{4}\left(10 + V_1\right) \quad ③$$

$V_2 \approx 5,73kV$
$V_1 \approx 2,19kV$
$V_3 \approx 3,05kV$

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a) Ionization by Collision (5 pts)

- A free electron hits a neutral atom.
- This creates one more electron and one positive ion.
- It happens when the electron has enough energy from the electric field.
- Common in gas discharges.

b) Photo-ionization (5 pts)

- A photon hits a neutral atom.
- If photon energy is high enough, it removes an electron.
- This helps start the ionization process, even if no free electrons exist at first.

c) Thermal Ionization (5 pts)

- High temperature gives energy to atoms.
- The atom loses an electron and becomes an ion.
- Seen in hot gases and plasmas.

d) Surface Ionization (10 pts)

1. Thermionic Emission:

- A hot surface releases electrons.
- Depends on temperature.

2. Field Emission:

- Strong electric field pulls electrons from the surface.
- Happens through quantum tunneling.

3. Secondary Emission:

- A particle hits the surface and knocks out electrons.

4. Photoelectric Emission:

- A light (photon) hits the surface and ejects electrons.
- Based on Einstein's photoelectric effect.

e) Electrode Gap – Breakdown Voltage Graph (5 pts)

- Draw a graph showing breakdown voltage vs. gap distance.
- Use 3 electrode types:
  1. Plane–Plane
  2. Point–Plane
  3. Sphere–Sphere
- Point–Plane has the lowest breakdown voltage.
- Curve looks like a Paschen curve.

Three tests presented by project groups:

1. Impulse Voltage Test
2. [redacted] Ground fault testing
3. [redacted] switchgear testing

Impulse Voltage Test (explained – not my group's test):

Impulse voltage test is used to check if electrical equipment can survive very high and short voltage pulses, like lightning.

In this test, a sharp high voltage is applied for a short time. The typical shape is 1.2/50 µs:

- 1.2 µs is the time to reach peak
- 50 µs is the time to drop to half

This test helps to make sure the insulation does not break under sudden voltage. It is very important for safety in power systems.